

# Comparative Evaluation of YOLOv8 Models for Food Object Detection

Viktorija Projkova, David Naumovski, Mihajlo Naumoski, Hristijan Gjoreski

Faculty of Electrical Engineering and Information Technologies,

University of Ss. Cyril and Methodius in Skopje, Macedonia

projkova.viktorija@gmail.com, davidnaumovski38@gmail.com, mihajlonaumoski07@gmail.com, hristijang@feit.ukim.edu.mk

**Abstract**—Wearable smart glasses that automatically capture meal images during eating events require accurate, real time, and privacy preserving processing. To minimize exposure of sensitive visual content, food regions should be detected on device, while only the relevant cropped content is forwarded for downstream nutritional analysis. This paper evaluates YOLOv8 based object detection for food localization with a focus on deployment on resource constrained hardware by comparing three model variants: YOLOv8n, YOLOv8s, and YOLOv8m. The models, pre trained on the MS COCO dataset, were fine tuned via transfer learning on a diverse food recognition dataset containing approximately 40k annotated images. All experiments were conducted under consistent conditions using the Ultralytics training pipeline on Google Colab Pro with NVIDIA GPU acceleration. Performance was evaluated quantitatively on labeled validation data using precision, recall, and IoU based mean average precision (mAP) metrics, and further assessed qualitatively on an unlabeled test split and real world images captured with Emteq smart glasses to validate generalization to unseen samples. The results highlight the accuracy efficiency trade off across model sizes: YOLOv8s and YOLOv8m achieve more stable and precise localization, whereas YOLOv8n demonstrates competitive performance with the fastest inference, making it the most suitable candidate for real time deployment on wearable hardware. Finally, a prototype pipeline is described, demonstrating how segmented food regions can be transmitted from a mobile application to a local vision language model backend (LLaVA) for meal classification and calorie estimation.

**Keywords**—YOLOv8; Food detection; Wearable Smart glasses; Mobile application; LLaVA;

## I. INTRODUCTION

Wearable technologies are increasingly used in health oriented applications, including automatic monitoring of daily nutrition. Smart glasses and other camera based wearables can capture images during eating events and support meal logging, food recognition, and calorie estimation. However, images collected in real world environments rarely contain only the meal. They often include sensitive background content such as faces, personal objects, and private surroundings, which introduces a major privacy concern. For this reason, full images should not be directly transmitted for processing. Instead, food regions should be localized first, so that subsequent analysis is performed only on the relevant cropped content. This task requires object localization rather than simple image classification. While standard CNN classification models can identify food categories, they do not provide precise information about where food appears in the image, which is necessary for cropping or masking background content.

Therefore, real time object detection becomes a more appropriate approach, especially under resource constraints typical for wearable devices.

Among real time object detectors, the YOLO (You Only Look Once) family is widely adopted due to its single stage design, which predicts bounding boxes and classes in one forward pass [1]. This architecture provides a strong balance between speed and accuracy, making YOLO suitable for mobile and wearable deployment scenarios. The Ultralytics YOLOv8 framework additionally offers multiple model sizes, enabling comparison of detection quality against computational complexity [3].

In this work, three YOLOv8 variants are compared: YOLOv8n, YOLOv8s, and YOLOv8m using a consistent transfer learning pipeline. The models were initialized from MS COCO pre trained weights and fine tuned on a diverse food recognition dataset containing approximately 40k annotated images. Performance was evaluated quantitatively on labeled validation data using precision, recall, and IoU based mAP metrics, and qualitatively through bounding box inspection on an unlabeled test split and external images captured with Emteq smart glasses to assess generalization to unseen real world conditions.

Besides comparing YOLOv8 variants, this work outlines a prototype workflow showing how localized food regions can be separated from the background and sent for further analysis. A local LLaVA based backend is used as an example for food classification and calorie estimation.

## II. RELATED WORK

Recent research in computer vision has shown strong progress in food image understanding, primarily through deep learning approaches. Many early studies focus on food classification, where the goal is to assign one or more food labels to an image. However, real world meal images are often complex scenes containing multiple objects and distracting background elements, which makes object detection a more suitable formulation. Large scale benchmarks and datasets, such as Food Recognition challenges, emphasize this transition from simple recognition to localization tasks, where models must correctly detect food items under varying viewpoints, occlusions, and real life conditions [4, 10].

Among object detection methods, the YOLO (You Only Look Once) family is widely adopted due to its single stage architecture, predicting bounding boxes and classes in one forward pass, and therefore providing high inference speed. Review studies on YOLOv8 highlight its efficiency, strong

performance, and the availability of multiple model sizes (n/s/m/l/x), enabling flexible deployment depending on hardware constraints and real time requirements [3]. In addition, comparative evaluations of YOLO versions across different settings confirm the importance of selecting the appropriate model size and configuration according to the use case, especially when balancing accuracy and computational cost [7].

Vora and Shekhawat present a comprehensive study on applying YOLO based object detection for food image recognition, motivated by the increasing demand for food related applications such as dietary tracking, restaurant menu analysis, and nutritional monitoring. Their work highlights YOLO as a suitable approach due to its real time processing capability and its ability to detect multiple objects in a single pass, which is important in complex food scenes with different food appearances, portion sizes, and occlusions. In addition, the authors discuss how pretrained YOLO models can be optimized through fine tuning on food specific datasets, and they analyze the impact of training strategies and data augmentation on recognition performance. The study also emphasizes practical use cases such as calorie estimation and meal planning, providing a general framework for building YOLO based food recognition systems [5].

The existing literature confirms the suitability of YOLO based detectors for real time food localization, yet there is limited focus on practical deployment constraints and privacy sensitive processing. The novelty of this work lies in a controlled comparison of YOLOv8n/s/m variants for food detection under a unified training pipeline, combined with evaluation on both dataset test images and real world smart glasses samples. This provides practical guidance for selecting a model that balances detection accuracy, inference speed, and privacy requirements in wearable, mobile nutrition monitoring applications.

### III. DATA

This study uses a publicly available, pre-annotated dataset for food object detection obtained from Kaggle (Food Recognition 2022) [6]. The dataset consists exclusively of food related images, covering a wide variety of meals and items, including breakfast dishes, main courses, snacks, desserts, and beverages. Such diversity increases the realism of the dataset and supports training models that can generalize across different food appearances, plating styles, and scene contexts.

The dataset is provided with an existing split into training, validation, and test subsets, allowing consistent experimentation without additional manual labeling. The annotated training and validation images include bounding boxes for food items, which makes the dataset suitable for supervised training of object detection models.

The training subset contains 39000 annotated images and was used to fine tune the YOLOv8 models through transfer learning. The validation subset contains 1000 annotated images and was used for quantitative evaluation and comparison between models. This split was used to monitor and compare metrics such as mean average precision (mAP), precision,

recall, and loss values across all YOLOv8 variants. The test subset includes 3000 images, but without ground truth annotations. Therefore, it was used for qualitative analysis, where predicted bounding boxes were visually inspected to assess how the models behave on unseen images from the same dataset distribution.

In addition to the Kaggle dataset, an external test set of approximately 800 images was collected using Emteq smart glasses. These real world images differ from the online dataset in both quality and scene complexity: lighting conditions are often darker, the overall image quality is lower, and the background typically contains more objects and clutter. For this reason, the external test set represents a significantly more challenging evaluation scenario and provides a realistic indication of whether the trained models can generalize wearable camera conditions and reliably localize food in practical environments.

Overall, the combination of a large scale annotated online dataset, and an external wearable camera test set enables evaluation under both controlled and realistic conditions. The Kaggle dataset provides sufficient labeled data for stable model training and reliable quantitative validation, while the unlabeled test images and Emteq samples allow qualitative inspection of model behavior on unseen scenes. This dataset design supports the main goal of the study, assessing whether YOLOv8 models can robustly localize food in diverse environments and serve as a reliable privacy preserving filtering step before further high level nutritional analysis.

### IV. METHODS

In this work, food identification is formulated as an object localization task, where the goal is to predict bounding boxes around food items in an image. For this purpose, the YOLOv8 family of detectors was selected due to its efficient single stage design and suitability for real time processing. Importantly, YOLOv8 is used only to localize food regions rather than to perform detailed food type classification. This localization stage is essential in privacy sensitive wearable scenarios, since it enables filtering of irrelevant or sensitive background content by isolating only the food region before any further processing.

To obtain robust detection performance, the models were trained using transfer learning. Three YOLOv8 variants (YOLOv8n, YOLOv8s, YOLOv8m) were initialized from MS COCO pre-trained weights and then fine-tuned on a food only dataset. The COCO dataset is widely used for scene understanding and contains complex images with a large set of object categories. Prior work highlights COCO as a large scale benchmark with diverse objects and contexts, allowing detectors to learn strong general visual representations and robust object localization behavior [2]. By leveraging this pre-training, the models preserve general knowledge of non food objects and background structure, while fine tuning specializes them for reliable localization of diverse meals and beverages.

The Food Recognition dataset used for training was originally stored in a Deep Lake chunk-based format. To enable training in the Ultralytics YOLOv8 pipeline, the dataset was converted into the standard YOLO detection format, including

image folders, bounding box label text files, and a YAML configuration file describing dataset splits. During training, Ultralytics automatically evaluated the models on the validation set and saved checkpoints. For each variant, the best checkpoint was selected according to validation performance and used for further analysis.

Model evaluation was conducted in two stages. Firstly, performance was assessed quantitatively on the labeled validation split using standard detection metrics. Secondly, qualitative evaluation was performed through visual inspection of predicted bounding boxes on the unlabeled Kaggle test split and on an external test set of images collected with Emteq smart glasses. The external images represent more challenging real world wearable conditions (darker lighting, lower quality, and background clutter), providing a realistic assessment of model robustness and generalization.

## V. EXPERIMENTAL SETUP

All experiments were conducted under a controlled setup to enable a fair comparison between YOLOv8 model sizes. Three YOLOv8 variants (YOLOv8n, YOLOv8s, and YOLOv8m) were fine-tuned using the same Ultralytics training pipeline and identical dataset splits. Training was executed on Google Colab Pro with NVIDIA A100 GPU acceleration, ensuring stable training performance and comparable runtime conditions. Under this setup, the evaluation primarily reflects the impact of model scale on detection accuracy and generalization.

A key component of the workflow was preparing the dataset in a format compatible with YOLO training. The original dataset was stored in Deep Lake format, organized in chunk-based structure. To train YOLOv8 models, the dataset was exported into the standard YOLO directory layout (train/validation/test). During export, each sample was decoded into an image file, and annotations from the labeled subsets were converted into YOLO label files. In YOLO format, each object instance is described as a single line:

$$(class_{id}, x_c, y_c, w, h) \quad (1)$$

Where  $x_c, y_c$  represent the center of the bounding box and  $w, h$  its width and height, all normalized by the image dimensions. A dataset YAML configuration file was generated to define paths to the splits and class information required by the Ultralytics framework. This conversion ensured consistent dataset preparation for all YOLOv8 variants.

Model performance was evaluated quantitatively on the labeled validation split using standard object detection metrics. The overlap between a predicted bounding box  $B_p$  and the ground-truth box  $B_{gt}$  is measured using Intersection over Union (IoU):

$$IoU = \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|} \quad (2)$$

A detection is considered a True Positive (TP) when the predicted class matches the ground truth and  $IoU \geq \tau$  for a selected IoU threshold  $\tau$ . False Positives (FP) correspond to incorrect detections or insufficient overlap, while False

Negatives (FN) represent missed objects. Precision and recall are defined as:

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

To summarize performance across confidence thresholds, Average Precision (AP) is computed as the area under the Precision-Recall curve for each class -  $k$ , while mean Average Precision (mAP) is obtained by averaging AP across all classes -  $n$ .

$$mAP = \frac{1}{n} \sum_{k=1}^n AP_k \quad (5)$$

In this study,  $mAP@0.5$  reports AP at  $\tau = 0.5$ , whereas  $mAP@0.5:0.95$  follows the COCO style evaluation by averaging AP over multiple IoU thresholds  $\tau \in \{0.50, 0.55, \dots, 0.95\}$ , thus providing a stricter measure of localization quality [2].

In addition to these metrics, YOLOv8 assigns a confidence score to each predicted bounding box, reflecting the model's certainty in its detection. Since the target class in our task is food, confidence can be interpreted as the likelihood that the detected region corresponds to food rather than background content. This score is also practically important for filtering detections, since adjusting the confidence threshold directly affects the precision-recall trade-off.

## VI. EXPERIMENTAL RESULTS

Before presenting the results, it is important to emphasize that the purpose of this experimental evaluation is not only to report detection accuracy, but primarily to determine which YOLOv8 model size provides the most suitable balance between localization quality and deployment feasibility. Since the intended application involves privacy preserving food detection in wearable and mobile settings, the models must consistently identify food regions under diverse conditions while remaining lightweight enough for real-time inference. For this reason, the evaluation was performed through a combination of quantitative validation metrics, to enable objective comparison, and qualitative visual inspection on unlabeled test data and real world images captured with Emteq smart glasses, to assess generalization and robustness.

Quantitative results were obtained using the labeled validation subset, where all YOLOv8 variants were compared under identical training conditions to isolate the impact of model scale. In terms of model complexity, YOLOv8n contains approximately 3.2M parameters, YOLOv8s 11.2M, and YOLOv8m 25.9M, which directly affects inference efficiency and suitability for deployment on wearable and mobile hardware. As summarized in Table 1, YOLOv8m achieved the strongest overall validation performance with  $mAP@0.5 = 0.7669$  and  $mAP@0.5:0.95 = 0.5093$ , together with the highest precision (0.8177) and recall (0.6346). YOLOv8s performed comparably, reaching  $mAP@0.5 = 0.7629$  and  $mAP@0.5:0.95 = 0.5060$ , with precision (0.8039) and recall (0.6385).

Finally, the lightweight YOLOv8n model achieved  $mAP@0.5 = 0.7474$  and  $mAP@0.5:0.95 = 0.4923$ , while

maintaining strong precision (0.7934) and recall (0.6321). Although the medium and small variants provide slightly higher mAP values and more stable localization, the overall quantitative gap remains relatively small, indicating that model efficiency can be improved significantly without major loss in detection quality.

Table 1. Best Epoch Validation Metrics for YOLOv8 Variants on the Food Dataset

|   | model   | best_epoch | best_map50_95 | best_map50 | best_precision | best_recall | best_box_loss | best_cls_loss | best_dfl_loss |
|---|---------|------------|---------------|------------|----------------|-------------|---------------|---------------|---------------|
| 2 | yolov8m | 20.0       | 0.50925       | 0.76693    | 0.81773        | 0.63457     | 0.99406       | 0.82882       | 1.61483       |
| 1 | yolov8s | 19.0       | 0.50600       | 0.76290    | 0.80394        | 0.63845     | 1.03216       | 0.88403       | 1.64503       |
| 0 | yolov8n | 20.0       | 0.49232       | 0.74741    | 0.79344        | 0.63213     | 1.04220       | 0.91223       | 1.65341       |

Since the Kaggle test split and the external Emteq dataset do not provide ground truth annotations, generalization performance was assessed qualitatively through visual inspection. To ensure a clear and consistent comparison, qualitative examples are presented in pairs for each model variant, showing detections on an unseen Kaggle test image and on a real world wearable image captured with Emteq smart glasses. Specifically, Figures 1-2 illustrate YOLOv8n predictions on Kaggle test and Emteq data, Figures 3-4 show corresponding YOLOv8s outputs, while Figures 5-6 provide qualitative results for YOLOv8m.



Figure 1 Qualitative comparison on the Kaggle test split: (a) YOLOv8n, (b) YOLOv8s, (c) YOLOv8m.



Figure 2 Qualitative comparison on real world images captured with Emteq smart glasses: (a) YOLOv8n, (b) YOLOv8s, (c) YOLOv8m.

Across the presented qualitative examples, YOLOv8s and YOLOv8m consistently produce tighter bounding boxes that align primarily with the food region, leading to more stable and precise localization. In comparison, YOLOv8n occasionally captures additional contextual elements such as parts of the plate, bowl, or container. Within the proposed privacy preserving pipeline, this behavior does not necessarily constitute a critical limitation, as limited contextual information may support downstream nutritional estimation and reduce ambiguity in food type recognition. Finally, for improved interpretability, each detection is accompanied by a confidence score, reflecting the model’s certainty when labeling a region as food and enabling a more transparent assessment of prediction reliability.

## VII. REAL-WORLD DEPLOYMENT AND QUALITATIVE EVALUATION

Testing the model under real-world conditions is essential for a meaningful qualitative evaluation, particularly in the context of wearable and mobile applications. For this reason, the trained YOLOv8 model was integrated into a prototype Android mobile application, where its sole purpose is to detect and localize food in images captured directly by the user [3].

The application was developed in Kotlin and implements a simple, user-driven workflow. First, the user captures an image using the device camera, which is then displayed in a preview screen. From this point, the user completes two explicit steps through separate button actions. The first button runs the YOLOv8 model, which was exported to TensorFlow Lite (TFLite) format to ensure compatibility and efficient inference on Android devices. Based on the model’s output, the



application overlays a bounding box corresponding to the detected food region [3, 9].

In the second step, the user activates a masking function that blackens all pixels outside the detected bounding box. This operation is intentionally designed to remove background content, such as faces or personal surroundings, thereby addressing privacy concerns before any image data is transmitted externally. Once masking is completed, the user can upload the processed image to a backend server for further analysis.

The backend server plays a central role in completing the proposed pipeline and enabling practical daily use. It is implemented in Python and integrates several supporting libraries for image processing and model inference. Pillow is used for image loading and resizing, while FastAPI provides a lightweight REST interface for communication between the mobile application and the server. For multimodal analysis, Ollama is used as a local inference runtime, hosting a LLaVA model from the LLaMA family. The masked image is passed to LLaVA, which performs food classification and provides a rough calorie estimate based on visual content [8].

The backend returns the analysis results to the mobile application in the form of a JSON object, containing the predicted food category and an estimated calorie value. The application then logs this information locally, allowing the user to maintain a daily record of consumed food items and total calorie intake. While the calorie estimates are not intended to be medically precise, this prototype demonstrates how reliable food localization can serve as a privacy-preserving preprocessing step for higher-level nutritional analysis.

## VIII. CONCLUSION

This work presented a comprehensive comparative evaluation of three YOLOv8 model variants—YOLOv8n, YOLOv8s, and YOLOv8m—for food object detection in the context of privacy-preserving wearable and mobile applications. Using a unified transfer-learning pipeline and consistent experimental conditions, the study demonstrated that all evaluated models are capable of reliably localizing food regions across diverse image sources, including curated online datasets and challenging real-world images captured with smart glasses.

Quantitative evaluation on labeled validation data showed that YOLOv8m and YOLOv8s achieve the highest detection accuracy, reflected in superior mAP, precision, and recall values, while YOLOv8n provides only a modest reduction in performance despite a significantly smaller parameter count. This finding highlights that lightweight detectors can maintain competitive localization quality, which is critical for real-time inference on resource-constrained wearable hardware. Qualitative inspection further confirmed that larger models produce tighter and more stable bounding boxes, whereas the nano variant occasionally includes additional contextual regions, a trade-off that may still be acceptable—or even beneficial—for downstream nutritional interpretation.

Beyond model comparison, this study demonstrated the practical feasibility of integrating food localization into a complete privacy-aware pipeline. By performing on-device detection and masking of background content prior to data

transmission, the proposed approach directly addresses privacy risks associated with continuous image capture in real-world environments. The successful integration with a local multimodal vision–language backend further illustrates how accurate object localization can serve as an effective preprocessing stage for higher-level tasks such as food classification and calorie estimation.

Overall, the results indicate that YOLOv8n offers the most favorable balance between accuracy, speed, and deployment efficiency for wearable scenarios, while YOLOv8s and YOLOv8m remain strong candidates when computational resources allow. Future work will focus on extending evaluation to a broader range of wearable devices, improving robustness under extreme lighting and occlusion conditions, and exploring adaptive confidence-based filtering strategies. Additionally, incorporating temporal information from video streams and expanding multimodal reasoning capabilities may further enhance the reliability and usability of privacy-preserving dietary monitoring systems.

## REFERENCES

- [1] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You Only Look Once: Unified, Real-Time Object Detection,” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA, 2016, pp. 779–788. <https://doi.org/10.1109/CVPR.2016.91>
- [2] T.-Y. Lin, M. Maire, S. Belongie, et al., “Microsoft COCO: Common Objects in Context,” *European Conference on Computer Vision (ECCV)*, Zurich, Switzerland, 2014, pp. 740–755. [https://doi.org/10.1007/978-3-319-10602-1\\_48](https://doi.org/10.1007/978-3-319-10602-1_48)
- [3] G. Jocher, A. Chaurasia, and J. Qiu, “YOLOv8 by Ultralytics,” GitHub repository, 2023. Available: <https://github.com/ultralytics/ultralytics>
- [4] S. P. Mohanty, D. P. Hughes, and M. Salathé, “Using Deep Learning for Image-Based Plant Disease Detection,” *Frontiers in Nutrition*, vol. 9, 2022. <https://doi.org/10.3389/fnut.2022.878628>
- [5] S. Vora and R. Shekhawat, “Deep Learning-Based Food Image Recognition Using YOLO,” *International Journal of Food and Nutritional Sciences (IJFANS)*, vol. 11, no. 3, 2022.
- [6] S. N. Reddy, “Food Recognition 2022 Dataset,” Kaggle, 2022. Available: <https://www.kaggle.com/datasets/sainikhileshreddy/food-recognition-2022>
- [7] T. H. Nguyen, H. T. Nguyen, and T. T. Le, “A Comparative Study of YOLOv8 Models for Object Detection,” *AI*, vol. 5, no. 4, 2024. <https://doi.org/10.3390/ai5040122>
- [8] H. Liu, C. Li, Q. Wu, and Y. J. Lee, “Visual Instruction Tuning,” *arXiv preprint*, arXiv:2304.08485, 2023. Available: <https://arxiv.org/abs/2304.08485>
- [9] TensorFlow Team, “TensorFlow Lite: Deploying Machine Learning Models on Mobile and Embedded Devices,” Google Developers Documentation, 2023. Available: <https://www.tensorflow.org/lite>
- [10] W. Min, S. Jiang, L. Liu, Y. Rui, and R. Jain, “A Survey on Food Computing,” *ACM Computing Surveys*, vol. 52, no. 5, 2019. <https://doi.org/10.1145/3321408>